Length Scale Analysis

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Document the length analysis of the system.

Characteristic distances of the system

The building blocks of the network are the interactions between patchy particles. From this two elements, the *reference* length is defined. The central particle has a radius of $1/2$ length units ($\sigma_C/2$), meanwhile the patches have a radius of $1/5$ length units ($\sigma_P/2$). The distance between the center of those particles is $2/5$ (h_P). On the other hand, the distance range of the interaction (r_b) between patches is σ_P length units and the cutoff distance is $3/2\sigma_P$ length units. With these distances we can define the distance of interaction between two patchy particles,

$$h_\mathrm{P} + r_\mathrm{b} + h_\mathrm{P} = 2\sigma_P + r_\mathrm{b}.$$

This distance is going to be refer as r_int and has the following range $\in [6/5, 7/5]$. Then, if we increase the amount of patchy particles in pair-wise interaction, the distance between central particles is given by

$$d_\mathrm{int} = (N-1)\left(2\sigma_p + r_b\right).$$